

Two Heads Are Better Than One: A Two-Stage Complex Spectral Mapping Approach for Monaural Speech Enhancement

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Abstract—For challenging acoustic scenarios as low signal-to-noise ratios, current speech enhancement systems usually suffer from performance bottleneck in extracting the target speech from the mixtures within one step. To address this issue, we propose a novel complex spectral mapping approach with a two-stage pipeline for monaural speech enhancement in the time-frequency domain. The proposed algorithm aims to decouple the primal problem into multiple sub-problems, which follows the classic proverb, “two heads are better than one”. More specifically, in the first stage, only magnitude is estimated, which is incorporated with the noisy phase to obtain a coarse complex spectrum estimation. To facilitate the previous estimation, in the second stage, an auxiliary network serves as the post-processing module, where residual noise is further suppressed and the phase information is effectively modified. The global residual connection strategy is adopted in the second stage to accelerate the training convergence speed. To alleviate the parameter burden caused by the multi-stage pipeline, we propose a light-weight temporal convolutional module, which substantially decreases the trainable parameters and obtains even better objective performance over the original version. We conduct extensive experiments on three standard corpora, including WSJ0-SI84, DNS Challenge dataset, and Voice Bank + DEMAND dataset. Objective test results demonstrate that our proposed approach achieves state-of-the-art performance over previous advanced systems under various conditions. Meanwhile, subjective listening test results further validate the superiority of our proposed method in terms of subjective quality.

Index Terms—Monaural speech enhancement, complex domain, two-stage, global residual connection, phase estimation

I. INTRODUCTION

Noise is all around us. When we hear someone’s voice, the signals we perceive are often contaminated by various noises, such as car siren and dog barking. These environmental interference may heavily affect the usage of automatic speech recognition (ASR) [1], telecommunication systems [2], and hearing assistant devices [3] in real scenarios. To tackle this

problem, speech enhancement (SE) is proposed to extract useful speech information from noisy signals, and improve the speech quality and intelligibility [4]. Due to the lack of spatial information, monaural speech enhancement remains a challenging topic for a long time, especially under low signal-to-noise ratios (SNRs). In this paper, we focus on noise reduction when only one channel recording is available.

Many well-known traditional algorithms have been proposed in the past several decades, including spectral-subtraction [5], Wiener filtering [6], statistical-based methods [7], [8], and so forth. By dint of the powerful capability of deep neural networks (DNNs), SE has been formulated as a supervised problem in recent years [9]. In the time-frequency (T-F) domain, typical targets can be divided into two categories, namely masking based [10] and mapping based [11]. For the former, ideal binary mask (IBM) [12] or ideal ratio mask (IRM) [13], tries to explore the energy distribution relation between clean and noise components. For the latter, log-power spectrum (LPS) [11] or magnitude spectrum (MS) [14] is leveraged as the mapping target.

For a long time, DNN-based SE methods focus on recovering the magnitude information but leaving the phase unaltered [10], [11], [15], which is mainly derived from two reasons. For one thing, phase information shows little temporal and spectral regularity in the T-F domain. For another, it has been claimed that phase is not that important for human perception [16]. Nonetheless, increasing evidence reveals that when the phase is also incorporated into optimization, the subjective perception quality can be notably improved, especially under low SNR conditions [17]. To this end, a multitude of DNN based algorithms incorporating phase are proposed henceforth and can be roughly divided into two streams, namely, complex-domain based [18]–[20] and time-domain based [21]–[26]. For the first type, phase information lies in the relative relation between real and imaginary (RI) components implicitly. For example, Williamson *et al.* proposed to stack fully-connected (FC) layers to estimate the complex ratio mask (CRM), which are separately applied to RI parts of the spectrum to recover the magnitude and phase simultaneously. However, the nonlinear function is usually utilized to compress the target dynamic range, which implicitly hampers the network training. More recently, Tan *et al.* [19] proposed a complex spectral mapping method, where the RI components are directly mapped with the help of a convolutional recurrent network (CRN), and empirically better results are achieved than CRM.

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For the other type, both input and output are raw waveform samples, diverting around the direct phase estimation problem. For example, In [22], a generative adversarial strategy based SE network named SEGAN was proposed for end-to-end training, where the waveform mixtures are first denoised with a U-Net based generative module and then recognized by the discriminator module. However, the approach just shows slight advantages over the traditional Wiener baseline in objective measures. In [23], Luo *et al.* proposed the ConvTasNet, where multiple temporal convolutional modules (TCMs) are stacked as the separator module to distinguish different sources in a trainable latent space and consistent signal-distortion ratio (SDR) improvements can be achieved compared with previous T-F domain based approaches. Despite the impressive performance of TCMs in temporal sequence modeling, the parameter burden is still notable, which heavily hinders the usage in the edge devices. From our perspective, it seems more advantageous to suppress the noise in the complex T-F domain than direct operation in the time domain, and the reasons are two-fold. Firstly, the usage of Fourier transform basis with fixed coefficients effectively decomposes the noise and speech components into a specific space, where the feature patterns tend to be more distinguishable and thus easier for network training [27]. Secondly, one can see that in the Interspeech 2020 DNS challenge¹, overall better Mean Opinion Score (MOS) results were achieved for T-F based schemes than those in the time-domain, and we speculate that the operation in the T-F domain may be more conducive to subjective quality.

Inspired by curriculum learning concept [28], multi-stage mapping algorithms begin to thrive in the speech front-end tasks recently [29]–[34], where the original mapping problem can be decomposed into multiple simpler sub-problems and a better solution can be obtained in each stage progressively as a result. In [29], [30], SNR is viewed as the progressive criterion for multi-stage training, where the spectrum with an improved SNR is selected as the target in each stage. Therefore, the noises can be attenuated in a stage-wise manner. However, as an independent module is involved in each stage, the number of trainable parameters will increase linearly with the number of stages, which heavily aggravates the parameter burden. In [34], the *masking and inpainting* logic is leveraged with a two-stage network, where a coarse spectrum is first obtained by noise filtering, followed by a direct mapping approach for spectrum inpainting. However, phase optimization is not considered, which restricts the performance upper bound. Moreover, the two stages are separately trained, and the advantages of joint optimization are neglected accordingly.

To overcome the performance bottleneck of previous SE systems in the challenging scenarios, we propose a novel framework with a Complex spectral mapping based Two-Stage Network (CTS-Net). It consists of two major sub-networks, namely *Coarse Magnitude Estimation Network* (dubbed ME-Net) and *Complex Spectrum Refine Network* (dubbed CS-Net), where the two stages are jointly trained in an end-to-end fashion. In the first stage, we aim to estimate the spectral magnitude with ME-Net, which is then coupled

with the original noisy phase to obtain a coarse complex spectrum. In the second stage, given the original and coarse complex spectra as the input, CS-Net attempts to further refine the previous spectrum by estimating the spectral detail, which complements both magnitude and phase information simultaneously. With this aim, a global residual connection is introduced in the second stage, *i.e.*, instead of directly estimating the complex spectrum, the second network only needs to estimate the residual, which is directly added to the previous coarse complex spectrum. We illustrate the rationale behind the proposed paradigm from three aspects. Firstly, for previous complex spectral mapping approaches, both real and imaginary components are estimated with RI loss, while the phase is implicitly optimized, the magnitude may deviate from its optimal optimization path, leading to sub-optimal solutions [35], [36]. In this paper, a sequential optimization strategy is leveraged, *i.e.*, when the magnitude is coarsely obtained, it shrinks the solution space and offers guidance for subsequent phase optimization. Secondly, there exists implicit relation between magnitude and phase [37]. For example, given clean magnitude as the reference, phase information can be gradually recovered by iteratively switching from short-time Fourier transform (STFT) to inverse STFT (iSTFT). As a result, the pre-estimated magnitude can better facilitate the optimization for phase. Thirdly, under low SNRs, some residual noises still retain after the first stage, and meanwhile, some spectral details tend to get lost due to the network performance limitation [33], so further noise suppression can be implemented in the second stage, and some spectral details are also further repaired.

In this paper, TCMs are utilized in both two sub-networks for temporal sequence modeling. To further alleviate the parameter burden, we propose a light-weight TCM called squeezed TCM (S-TCM), which decreases the amount of parameters by around 72%. The innovations are two-fold. Firstly, instead of doubling the feature dimension with depthwise dilated convolutions in the original version, we adopt the regular dilated convolutions [38] to further squeeze the feature size while improving the transform capability. Secondly, motivated by gating mechanisms [14], we add the gated branch within the module, which better facilitates the information flow. Ablation studies show that the S-TCM exhibits comparable or even better performance than the original version despite tremendous parameters are eliminated.

We extend our preliminary study [39], [40] by providing extensive experiments with advanced baselines in both causal and non-causal settings. The main contribution of this paper is two-fold. Firstly, we propose a complex-domain based two-stage paradigm to effectively improve the speech quality in the challenging acoustic scenarios. Secondly, we propose a novel TCM to substantially alleviate the parameter burden over the original version. The experiments are conducted on three corpora, namely WSJ0-SI84, DNS Challenge dataset, and Voice Bank + DEMAND dataset. The experimental results show that our proposed approach yields state-of-the-art (SOTA) performance than previous advanced systems.

The rest of our paper is organized as follows. In Section II, we formulate the speech enhancement problem in the T-F

¹<https://dns-challenge.azurewebsites.net/phase1results/Interspeech2020>

domain. In Section III, the proposed algorithm framework and the corresponding modules are illustrated in detail. Section IV introduces the experimental setup. In Section V, experimental results and analysis are given. Section VI gives the discussions, and some conclusions are drawn in Section VII.

II. SIGNAL MODEL AND PROBLEM FORMULATION

A. Signal Model

In real environments, received noisy signal $x(t)$ in the time domain can be modeled as:

$$x(t) = s(t) + n(t), \quad (1)$$

where $s(t)$ and $n(t)$ denote clean and noise signals of the time index t . Given STFT, the signals in the time domain can be converted into T-F domain, given as:

$$X_{k,l} = S_{k,l} + N_{k,l}, \quad (2)$$

where $X_{k,l} = |X_{k,l}|e^{j\theta_{X_{k,l}}}$, $S_{k,l} = |S_{k,l}|e^{j\theta_{S_{k,l}}}$, and $N_{k,l} = |N_{k,l}|e^{j\theta_{N_{k,l}}}$ denote the complex spectrum representations with respect to noisy, clean and noise signals in the frequency index k and time frame index l . For facilitating notations, we omit the index (k, l) henceforth if no conflict arises.

B. Problem Formulation

In previous complex spectral mapping approaches [19], a single network is utilized to estimate RI components, *i.e.*,

$$\{\tilde{S}_r, \tilde{S}_i\} = \mathcal{G}_{si}(X_r, X_i; \Phi_{si}), \quad (3)$$

where $\mathcal{G}_{si}(\bullet; \Phi_{si})$ denotes the mapping function of a single network with parameter set Φ_{si} , and symbol $(\bullet)$ denotes the estimated variable. Subscript $(\bullet)_r$ and $(\bullet)_i$ denote the real and imaginary parts of a complex variable, respectively. Mean square error (MSE) is generally used as the objective loss to optimize RI, *i.e.*,

$$\mathcal{L}^{RI} = \left\| S_r - \tilde{S}_r \right\|_2^2 + \left\| S_i - \tilde{S}_i \right\|_2^2. \quad (4)$$

In [35], the magnitude constraint was introduced as the regularization term to mitigate the optimization deviation of the magnitude, which proves to be significant in the complex spectrum recovery. Based on that, the loss is modified as

$$\mathcal{L} = \alpha \cdot \mathcal{L}^{RI} + (1 - \alpha) \cdot \mathcal{L}^{Mag}, \quad (5)$$

$$\mathcal{L}^{Mag} = \left\| \sqrt{|\tilde{S}_r|^2 + |\tilde{S}_i|^2} - \sqrt{|S_r|^2 + |S_i|^2} \right\|_2^2, \quad (6)$$

where $\alpha \in [0, 1]$ is the weighted coefficient between RI loss and the magnitude constraint.

III. PROPOSED TWO-STAGE SYSTEM

The diagram of our proposed system is presented in Fig. 1. It consists of two sub-networks, namely ME-Net and CS-Net. The pipeline goes as follows. In the first stage, the network, *i.e.*, ME-Net, aims to estimate the magnitude, which is dubbed $|\tilde{S}^{me}|$. Then, we couple the estimated magnitude with the original noisy phase to obtain a coarse complex spectrum.

However, due to network estimation error and performance limitation, some residual noise components may preserve and meanwhile, the speech distortion may occur during the mapping process. To resolve this problem, the second network, *i.e.*, CS-Net, receives both original noisy spectrum and coarse estimation as the input and tries to capture a type of residual estimation, dubbed $\tilde{S}^{cs}$, which is directly added to the coarse estimation to obtain the final estimation. In a nutshell, the whole forward calculation process is formulated as:

$$|\tilde{S}^{me}| = \mathcal{G}_1(|X|; \Phi_1), \quad (7)$$

$$\tilde{S}_r^{me} = \Re\left(|\tilde{S}^{me}|e^{j\theta_X}\right), \tilde{S}_i^{me} = \Im\left(|\tilde{S}^{me}|e^{j\theta_X}\right), \quad (8)$$

$$\{\tilde{S}_r^{cs}, \tilde{S}_i^{cs}\} = \mathcal{G}_2(X_r, X_i, \tilde{S}_r^{me}, \tilde{S}_i^{me}; \Phi_2), \quad (9)$$

$$\tilde{S}_r = \tilde{S}_r^{me} + \tilde{S}_r^{cs}, \quad (10)$$

$$\tilde{S}_i = \tilde{S}_i^{me} + \tilde{S}_i^{cs}, \quad (11)$$

where $\Re$ and $\Im$ denote real and imaginary parts. $\mathcal{G}_1(\bullet; \Phi_1)$ and $\mathcal{G}_2(\bullet; \Phi_2)$ denote the mapping functions for the first and the second stages, respectively.

A. Dilated Convolutions

In speech-related tasks such as speech recognition or voice conversion, it is usually necessary and significant to capture the hidden information within the speech, *e.g.*, the sonic of the speaker and the phoneme patterns, which require both local and global contexts. To enlarge the receptive field efficiently, dilated convolutions were proposed to aggregate the contexts from different scales [38].

Take 2-dimensional (2-D) convolution as an example. Given $F(\bullet) \in \mathbb{Z}^2$ as the input feature map and $k(\bullet) \in [-r, r]$ as the discrete filter with size $(2r+1)^2$, the dilated convolution is calculated as

$$(F *_d k)(\mathbf{p}) = \sum_{\mathbf{s} + d\mathbf{t} = \mathbf{p}} F(\mathbf{s}) k(\mathbf{t}), \quad (12)$$

where $*_d$ denotes the convolution operator with dilated rate d . When $d = 1$, the operator will degrade to the regular convolution. Note that though 2-D case is illustrated for dilated operator herein, it also applies to 1-D case.

B. Gated Linear Units

Thanks to gating mechanisms, recurrent neural networks (RNNs) can better mitigate the gradient vanishing problem in the propagation of each timestep, and variant RNNs like long short-term memory units (LSTMs) help to establish long-term memory by setting multiple gates with different roles [41]. Motivated by that, Dauphin *et al.* [42] proposed the gated linear unit (GLU) by introducing a gating branch, which is shown as:

$$\mathbf{y} = (\mathbf{x} * \mathbf{W} + \mathbf{b}) \otimes \sigma(\mathbf{x} * \mathbf{V} + \mathbf{c}), \quad (13)$$

where $\mathbf{x}$ and $\mathbf{y}$ are input and output feature maps. $\mathbf{W}$, $\mathbf{V}$ denote the convolution filter weights and $\mathbf{b}$, $\mathbf{c}$ are biases. σ is to scale the value range into $(0, 1)$. In this way, the

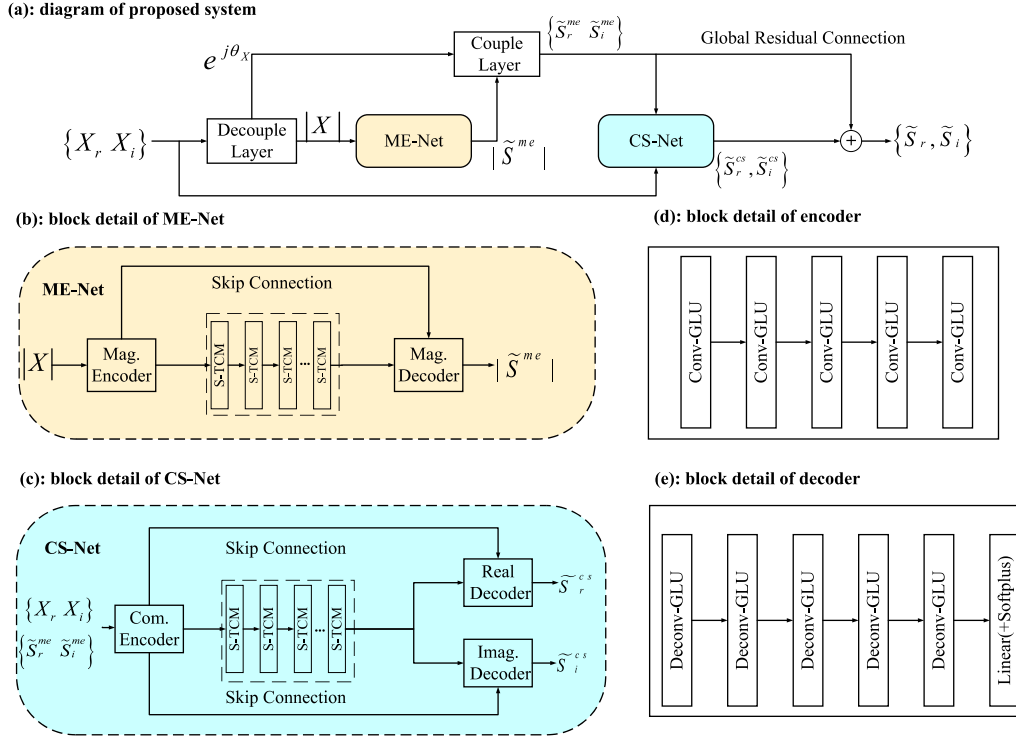


Fig. 1. (a): the diagram of the proposed CTS-Net. It consists of two sub-networks, namely ME-Net and CS-Net. ME-Net receives the noisy magnitude of spectrum and outputs the estimated counterpart, which is coupled with noisy phase to obtain a coarse spectrum. CS-Net receives both original noisy spectrum and coarse spectrum and outputs the complex residual, which is directly summated to the coarse complex spectrum. (b): the block detail of ME-Net. It consists of one encoder, decoder and stacked squeezed temporal convolutional modules (S-TCMs). (c): the block detail of CS-Net, which consists of one encoder, two decoders and stacked S-TCMs. (d): the block detail of encoder, which includes multiple Conv-GLU blocks. (e): the block detail of decoder, which includes multiple Deconv-GLU blocks.

information in the main branch can be dynamically controlled and the gradient information can also better propagate in the network [42], [43].

C. Squeezed Temporal Convolutional Modules

Recently, TCMs have been vastly applied in speech separation [23], [33], [44] and target speaker extraction tasks [45]. Compared with LSTMs, TCMs can obtain comparable even better performance than LSTMs in temporal sequence modeling and meanwhile, as parallelable convolutions are used, less inference time is needed. Fig. 2 (a) presents the architecture of TCM. It consists of three principal components, namely input $1 \times 1 - conv$, depthwise dilated convolution (dubbed $DD - Conv$) and output $1 \times 1 - conv$. Parametric ReLU [46] and normalization layer are inserted between adjacent convolutions. Given input feature with size (C_{in}, T) , where C_{in} and T refer to input channels and timesteps, respectively, after input convolution, the channel dimension is first doubled into a higher feature space, e.g., C_1 , followed by $DD - Conv$ to explore long-range sequence relation. Then output convolution switches the channel back to C_{in} . Residual connection is utilized to mitigate gradient vanishing problem.

Despite satisfactory performance has been made with s-tacked TCMs, the parameter burden problem is still noticeable. Specifically, although the depthwise convolution brings negligible parameters compared with regular convolution, as the channel dimension is doubled, the major burden falls into the input and output convolutions, i.e., $2C_{in}C_1 + K_1C_1 \approx$

$2C_{in}C_1, C_{in} \gg K_1$, where K_1 is the kernel size. In addition, the depthwise convolution shows less transform capability than the regular convolution due to its independence characteristic in each channel.

To further ease the parameter burden, we propose a lightweight sequential modeling unit called squeezed TCM (S-TCM), as shown in Fig. 2 (b). The innovations are two-fold. Firstly, different from doubling the feature dimension with $DD - Conv$, we squeeze the feature into a lower dimension C_2 with a regular dilated convolution. The rationale is that the spectrum usually exhibits very sparse distribution in the T-F domain, and useful information can be well preserved with low-dimension representations. Secondly, inspired by GLUs, apart from the main dilated convolution (main branch), we also add another branch called gated branch, which has a similar structure as the main branch except the sigmoid function is used to modulate the values into $(0, 1)$. In this way, the information can better flow in the gradient propagation.

Now we analyze the parameter superiority of S-TCM over TCM. As the number of parameters with respect to activation functions and norm layers are usually negligible than convolutions, we omit them in the parameter calculation for analysis convenience. For TCM, the number of parameters Ω_1 gives as:

$$\Omega_1 = 2C_{in}C_1 + K_1C_1, \quad (14)$$

and for S-TCM,

$$\Omega_2 = 2C_{in}C_2 + 2K_2C_2^2, \quad (15)$$

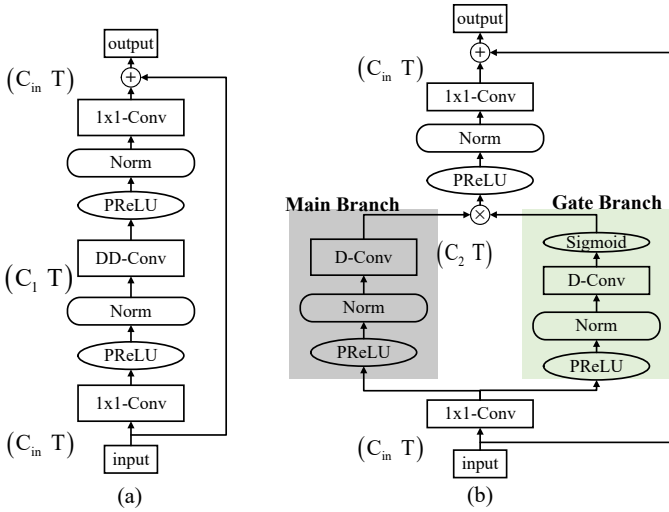


Fig. 2. (a): illustration of TCM. (b): illustration of the proposed squeezed TCM (S-TCM).

where K_2 is the kernel size of S-TCM. Assuming $C_1 = pC_2$, $C_{in} = qC_2$, $\{p, q\} \in \mathbb{R}^+$, then the parameter gain η gives as:

$$\eta = \frac{\Omega_1}{\Omega_2} = \frac{2pqC_2 + K_1p}{2(q + K_2)C_2}, \quad (16)$$

Consider the case when $C_{in} = 256$, $C_1 = 512$, $K_1 = 3$, which is typical for TCM setup [23]. $C_2 = 64$, $K_2 = 5$ is chosen for S-TCM setup in this study. Therefore, $p = 8$ and $q = 4$, i.e., compressed feature has much smaller than the doubled dimension size in S-TCM, then $\eta \approx 3.58$, indicating proposed S-TCM can reduce the parameter burden by around 72% than TCM. Note that despite the number of parameters is dramatically decreased, S-TCM still exhibits comparable or even better performance over TCM, which will be analyzed in Sec.V-D.

D. Proposed Framework Details

The network details of ME-Net and CS-Net are presented in Fig. 1 (b)-(c). One can find that they have the similar topology, except one decoder in ME-Net is for magnitude estimation while two decoders in CS-Net are utilized for RI estimation. Note that for ME-Net, as the target is the magnitude spectrum $|\tilde{S}^{me}| \in [0, +\infty)$, Softplus [47] is used as the output activation function. For CS-Net, as both real and imaginary spectra have unbounded ranges, we use the linear outputs. For effective temporal sequence modeling, stacked S-TCMs are inserted in the bottleneck, which gradually grasp the long-range temporal dependencies.

1) *Encoder and Decoder Modules*: The details of the encoder and decoder are shown in Fig. 1 (d)-(e). Note that different from using simple 2-D convolution layers in the encoder and decoder modules, we adopt the GLU formats, and the corresponding blocks are called Conv-GLU and Deconv-GLU blocks, respectively. For the encoder, five Conv-GLU blocks are utilized to consecutively compress the spectral features while capturing the local spectral-temporal patterns. Within each block, the kernel size is set to $(2, 3)$ in the time and frequency axis except $(2, 5)$ in the first block. The stride is

$(1, 2)$, which means that while the frequency size is gradually halved, the time size remains unchanged to meet the real-time requirement. Different from the setup in [48] that the number of channels exponentially increases from 16 to 256, the channel size in each layer is set to 64 since we empirically found that more channels in the first several layers could facilitate the performance than that in the last several layers. After each convolution, InstanceNorm (IN) [49] and PReLU layers are followed to accelerate the training. The decoder is the mirror version of the encoder, except all the Conv-GLUs are replaced by Deconv-GLUs, where the compressed features are gradually interpolated and restored to the original size. To mitigate the information loss during the encoding process, the skip connection strategy is utilized in both networks.

2) *Temporal Sequence Modeling Module*: For both ME-Net and CS-Net, we use 3 groups of S-TCMs as the sequential module, each of which stacks 6 S-TCM units with dilation rate exponentially increasing to obtain a large temporal receptive field, i.e., $(1, 2, 4, 8, 16, 32)$. In this way, the network can leverage the relation among different temporal scales to boost the speech information recovery. When no future frames are involved into current frame inference, the module becomes causal and vice versa. Note that despite CS-Net adopts the similar network topology to ME-Net in this paper, some advanced modules like transformer [50] can also be utilized to better capture the detailed information, which is out of the scope of this paper.

As both ME-Net and CS-Net have the similar structure, we only take ME-Net as an example, and the detailed parameter setup for ME-Net is presented in Table I. The input-size is specified with $(Timesetep \times FeatureSize)$ for 2-D format and $(ChannelNum \times Timesetep \times FeatureSize)$ for 3-D format. For the encoder and the decoder, the hyperparameters are specified with $(KernelSize, Stride, ChannelNum)$ format for both encoder and decoder, and $(kernelSize, dilationRate, ChannelNum)$ for S-TCMs. In addition, δ refers to the number of channels of network input. For ME-Net, as only noisy magnitude spectrum is used, $\delta = 1$. For CS-Net, both original and previously estimated complex spectra are used, $\delta = 4$.

E. Loss Function

In our CTS-Net, we adopt a two-stage training pipeline as illustrated previously. First, we train ME-Net with MSE loss until convergence,

$$\mathcal{L}_{me} = \left\| |\tilde{S}^{me}| - |S| \right\|_F^2, \quad (17)$$

Then we train ME-Net and CS-Net jointly, and the overall loss is given as:

$$\mathcal{L} = \mathcal{L}_{cs} + \lambda \mathcal{L}_{me}, \quad (18)$$

where λ denotes the weighted coefficient between the two losses, and the loss in CS-Net is defined as:

$$\mathcal{L}_{cs} = \alpha \mathcal{L}_{cs}^{RI} + (1 - \alpha) \mathcal{L}_{cs}^{Mag}, \quad (19)$$

where

$$\mathcal{L}_{cs}^{RI} = \left\| \tilde{S}_r - S_r \right\|_F^2 + \left\| \tilde{S}_i - S_i \right\|_F^2, \quad (20)$$

TABLE I
DETAILED PARAMETER SETUP FOR ME-NET.

Encoder	layer name	input size	hyperparameters	output size
	convglu_1	$\delta \times T \times 161$	$2 \times 5, (1, 2), 64$	$64 \times T \times 79$
	convglu_2	$64 \times T \times 79$	$2 \times 3, (1, 2), 64$	$64 \times T \times 39$
	convglu_3	$64 \times T \times 39$	$2 \times 3, (1, 2), 64$	$64 \times T \times 19$
	convglu_4	$64 \times T \times 19$	$2 \times 3, (1, 2), 64$	$64 \times T \times 9$
	convglu_5	$64 \times T \times 9$	$2 \times 3, (1, 2), 64$	$64 \times T \times 4$
reshape_size_2		$64 \times T \times 4$	-	$T \times 256$
S-TCMs		$T \times 256$	$\left(\begin{array}{l} 1, 1, 64 \\ 5, 1, 64 \\ 1, 1, 256 \\ 1, 1, 64 \\ 5, 2, 64 \\ 1, 1, 256 \\ 1, 1, 64 \\ 5, 4, 64 \\ 1, 1, 256 \\ 1, 1, 64 \\ 5, 8, 64 \\ 1, 1, 256 \\ 1, 1, 64 \\ 5, 16, 64 \\ 1, 1, 256 \\ 1, 1, 64 \\ 5, 32, 64 \\ 1, 1, 256 \end{array} \right) \times 3$	$T \times 256$
reshape_size_3		$T \times 256$	-	$64 \times T \times 4$
Decoder	skip_1	$64 \times T \times 4$	-	$128 \times T \times 4$
	deconvglu_1	$128 \times T \times 4$	$2 \times 3, (1, 2), 64$	$64 \times T \times 9$
	skip_2	$64 \times T \times 9$	-	$128 \times T \times 9$
	deconvglu_2	$128 \times T \times 9$	$2 \times 3, (1, 2), 64$	$64 \times T \times 19$
	skip_3	$64 \times T \times 19$	-	$128 \times T \times 19$
	deconvglu_3	$128 \times T \times 19$	$2 \times 3, (1, 2), 64$	$64 \times T \times 39$
	skip_4	$64 \times T \times 39$	-	$128 \times T \times 39$
	deconvglu_4	$128 \times T \times 39$	$2 \times 3, (1, 2), 64$	$64 \times T \times 79$
	skip_5	$64 \times T \times 79$	-	$128 \times T \times 79$
	deconvglu_5	$128 \times T \times 79$	$2 \times 5, (1, 2), 64$	$1 \times T \times 161$
	reshape_size_4	$1 \times T \times 161$	-	$T \times 161$
linear_1(Softplus)		$T \times 161$	161	$T \times 161$

$$\mathcal{L}_{cs}^{Mag} = \left\| \sqrt{|\tilde{S}_r|^2 + |\tilde{S}_i|^2} - \sqrt{|S_r|^2 + |S_i|^2} \right\|_F^2. \quad (21)$$

We empirically find that $\alpha = 0.5$ and $\lambda = 0.1$ suffice in our evaluation.

IV. EXPERIMENTAL SETUP

A. Dataset

We conduct extensive experiments on the WSJ0-SI84 dataset [51] for model evaluations, which consists of 7138 utterances by 83 speakers (42 males and 41 females). 5428 and 957 utterances with 77 speakers are selected for training and model validation, respectively. Two types of test sets are provided, each of which includes 150 utterances by 6 speakers (3 males and 3 females). For the first type, the speaker information is within the training set (dubbed seen speaker). For the second type, the speaker information is untrained to validate the generalization capability of models toward different speakers (dubbed unseen speaker).

To generate the noisy-clean pairs, we randomly select around 20,000 environmental noises from the Interspeech 2020 DNS-Challenge dataset² as the noise set, whose duration is around 55 hours. During each mixed process, a random noise

cut is generated to obtain a noise vector, which is subsequently mixed with a randomly selected clean utterance under a SNR level. The SNR range for training is $[-5, 0]$ dB with the interval 1dB. As a result, we totally generate 150,000 and 10,000 noisy-clean pairs for training and validation, respectively. The total duration of the training set is around 300 hours. For model testing, we select two challenging noises from NOISEX92 [52], namely babble and factory1. Five SNRs are set, namely -6dB, -3dB, 0dB, 3dB and 6dB. 150 noisy-clean pairs are generated for each case.

B. Parameter Setup

All the utterances are sampled at 16kHz and chunked to 8 seconds for training stability. The 20ms Hanning window is utilized, with 50% overlap in adjacent frames. 320 point FFT is used, which leads to 161-D spectral features. The models are optimized by Adam ($\beta_1 = 0.9, \beta_2 = 0.999$) [53]. In the first stage, the initialized learning rate (LR) is set to 0.001. In the second stage, ME-Net is fine-tuned with LR = 0.0001, while 0.001 for CS-Net. The learning rate will be halved if consecutive three validation loss increment arises and the network will be early-stopped if that of five. 60 epochs are for network training in total, and the batch size is set to 8 at the utterance level.

C. Baselines

In this study, we compare our proposed model with various advanced baselines for both causal and non-causal systems. All the models are trained with 1 Nvidia Tesla V100 using Pytorch [54]. For causal settings, five baselines are selected, namely LSTM [55], CRN [48], GCRN [19], DCCRN [56], and ConvTasNet [23]. LSTM and CRN are based on magnitude-only estimation. GCRN is an advanced version of CRN, where RI components are estimated for phase recovery, and all the regular convolutions in the encoder and decoder are replaced by their GLU versions. DCCRN incorporates complex-value based convolutions and LSTMs together for complex spectrum recovery. ConvTasNet is an advanced end-to-end time-domain system for speaker separation, where different sources can be distinguished and separated in the learned over-complete feature space. For non-causal settings, six baselines are selected, namely BLSTM, BCRN, GRN [14], DCN [57], AECNN [58] and BGCRN. BLSTM, BCRN, and BGCRN are similar to that of LSTM, CRN, and GCRN, except all the LSTM layers are replaced by their bidirectional versions. GRN and DCN are both fully convolutional networks (FCNs), which incorporate dilated GLUs and residual connections for magnitude recovery. AECNN is a state-of-the-art time-domain based model, which utilizes a typical 1-D U-Net to reconstruct the waveform samples directly. All the baselines are trained with the best parameter configurations mentioned in the literature, except several modifications are set. Firstly, for both GCRN and BGCRN, except for RI loss, we also introduce the magnitude constraint for better objective performance. Secondly, for DCCRN, instead of using SI-SDR [59] as the loss, we utilize “RI+Mag.” loss for fair comparison among complex-domain based peers. Besides, semi-causal convolutions are replaced

²<https://github.com/microsoft/DNS-Challenge>

TABLE II

OBJECTIVE RESULT COMPARISONS AMONG DIFFERENT MODELS IN TERMS OF PESQ, ESTOI AND SDR FOR BABBLE AND FACTORY1 NOISES IN THE SEEN SPEAKER TEST SET. **CAU.** DENOTES WHETHER TO USE THE CAUSAL SETUP. **BOLD** INDICATS THE BEST SCORE IN EACH CASE. BOTH CAUSAL AND NON-CAUSAL SYSTEMS ARE PROVIDED.

Metrics		PESQ							ESTOI(%)							SDR(dB)						
SNR(dB)		-6	-3	0	3	6	Avg.		-6	-3	0	3	6	Avg.		-6	-3	0	3	6	Avg.	
Babble Noise	Noisy	✓	1.54	1.72	1.88	2.06	2.25	1.89	26.63	34.76	43.27	51.95	61.61	43.65		-5.90	-2.94	0.04	3.03	6.03	0.05	
	LSTM	✓	1.81	2.09	2.31	2.52	2.70	2.28	45.80	56.22	64.94	71.59	77.32	63.18		0.61	3.44	5.88	8.15	10.14	5.64	
	CRN	✓	1.82	2.14	2.40	2.64	2.87	2.37	48.24	58.48	67.21	73.87	79.84	65.53		1.10	4.03	6.63	8.92	11.25	6.39	
	GCRN	✓	2.03	2.37	2.65	2.89	3.10	2.61	53.92	64.16	72.32	78.14	82.58	70.23		3.50	6.23	8.59	10.50	12.19	8.20	
	DCCRN	✓	1.96	2.29	2.60	2.84	3.07	2.55	49.82	60.53	69.89	76.43	82.04	67.74		2.69	5.74	8.51	10.80	13.01	8.15	
	ConvTasNet	✓	1.97	2.29	2.57	2.78	3.00	2.52	57.13	67.23	75.06	80.57	84.95	72.99		4.68	7.34	9.85	12.01	13.99	9.57	
	ME-Net(Pro.)	✓	1.97	2.28	2.54	2.77	2.98	2.51	51.02	61.39	70.09	76.41	81.84	68.15		1.86	4.74	7.27	9.57	11.91	7.07	
	CTS-Net(Pro.)	✓	2.20	2.56	2.87	3.10	3.28	2.80	59.97	70.39	77.99	82.89	86.58	75.57		5.19	8.33	10.90	12.90	14.74	10.41	
	BLSTM	✗	2.09	2.41	2.68	2.90	3.09	2.63	57.09	67.20	75.22	80.26	84.39	72.83		2.17	4.86	7.33	9.53	11.66	7.11	
	BCRN	✗	2.07	2.38	2.64	2.84	3.04	2.59	55.96	66.05	73.67	79.10	83.63	71.68		2.09	4.90	7.36	9.52	11.78	7.13	
	GRN	✗	1.89	2.16	2.37	2.54	2.70	2.33	50.10	59.92	68.20	74.35	79.74	66.46		1.61	4.37	6.82	9.05	11.25	6.62	
	DCN	✗	1.86	2.15	2.41	2.62	2.82	2.37	47.15	57.38	66.26	72.99	78.59	64.47		1.34	4.16	6.75	9.03	11.21	6.50	
	AECNN	✗	1.90	2.30	2.63	2.90	3.13	2.57	53.32	64.84	73.49	79.62	84.14	71.08		4.18	7.16	9.84	11.96	13.95	9.42	
	BGCRN	✗	2.23	2.59	2.88	3.11	3.28	2.82	58.99	70.12	77.90	83.09	86.58	75.34		4.55	7.62	10.14	12.23	14.17	9.74	
Factory1 Noise	ME-Net(Pro.)	✗	2.04	2.37	2.63	2.85	3.06	2.59	54.69	65.16	73.15	78.67	83.44	71.02		2.05	4.99	7.52	9.80	12.11	7.29	
	CTS-Net(Pro.)	✗	2.37	2.76	3.05	3.24	3.39	2.96	65.05	74.94	81.33	85.22	88.23	78.95		6.15	9.16	11.61	13.46	15.14	11.10	
	Noisy	-	1.45	1.60	1.78	1.99	2.20	1.80	26.00	34.76	43.73	54.04	63.90	44.49		-5.89	-2.92	0.05	3.04	6.03	0.06	
	LSTM	✓	1.92	2.18	2.40	2.59	2.75	2.37	46.14	56.84	65.20	72.38	78.07	63.72		2.53	4.86	7.03	9.06	10.80	6.86	
	CRN	✓	1.94	2.23	2.48	2.71	2.92	2.46	46.54	57.69	66.39	74.30	80.48	65.08		2.51	5.11	7.44	9.74	11.88	7.34	
	GCRN	✓	2.13	2.45	2.72	2.93	3.13	2.67	51.70	63.34	71.39	77.73	82.58	69.35		4.45	6.92	8.99	10.73	12.34	8.69	
	DCCRN	✓	2.14	2.45	2.71	2.93	3.14	2.67	49.57	61.18	70.06	77.07	82.62	68.10		4.63	7.13	9.38	11.51	13.48	9.23	
	ConvTasNet	✓	2.09	2.36	2.61	2.83	3.03	2.58	55.48	65.99	73.83	80.07	84.66	72.01		5.80	8.19	10.37	12.40	14.28	10.21	
	ME-Net(Pro.)	✓	2.09	2.36	2.59	2.81	3.01	2.57	50.29	60.76	69.07	76.32	82.00	67.69		3.24	5.67	7.92	10.22	12.42	7.89	
	CTS-Net(Pro.)	✓	2.33	2.64	2.88	3.08	3.26	2.84	59.26	69.22	76.36	81.79	86.03	74.53		6.53	8.97	11.06	12.93	14.70	10.84	
	BLSTM	✗	2.27	2.53	2.75	2.96	3.14	2.73	58.91	68.40	75.18	80.54	84.78	73.56		3.65	5.89	8.03	10.20	12.19	7.99	
	BCRN	✗	2.21	2.48	2.70	2.90	3.07	2.67	55.88	66.10	73.16	79.12	83.87	71.63		3.32	5.73	7.96	10.21	12.31	7.91	
	GRN	✗	2.03	2.26	2.43	2.58	2.70	2.40	52.32	62.06	69.70	76.09	81.25	68.28		3.17	5.60	7.86	10.07	12.10	7.76	
	DCN	✗	2.05	2.32	2.55	2.74	2.92	2.52	48.42	59.47	68.05	74.67	80.18	66.16		3.11	5.60	7.85	10.02	12.03	7.72	
	AECNN	✗	2.06	2.44	2.72	2.97	3.16	2.67	52.02	64.12	72.74	78.86	83.54	70.26		5.70	8.14	10.26	12.14	13.92	10.03	
	BGCRN	✗	2.35	2.69	2.95	3.16	3.31	2.89	57.29	69.24	77.11	82.63	86.45	74.54		5.20	7.78	10.20	12.21	14.05	9.89	
	ME-Net(Pro.)	✗	2.18	2.45	2.68	2.90	3.09	2.66	54.84	64.76	72.45	78.64	83.58	70.85		3.41	6.00	8.14	10.42	12.61	8.11	
	CTS-Net(Pro.)	✗	2.49	2.80	3.03	3.21	3.36	2.98	64.71	73.90	80.02	84.27	87.60	78.10		7.28	9.60	11.56	13.33	14.99	11.35	

by the causal versions³. Thirdly, for AECNN, different from merely using the frequency loss in [58], we also add the time-domain loss as the multi-task learning and empirically better objective performance can be achieved. Besides, the frame size for input and output of AECNN is 16384 samples with 50% frame overlap, *i.e.*, around 1-second contexts can be leveraged for each frame. As ConvTasNet is originally utilized to separate signals under 8kHz sampling rate, we extend it to 16kHz for model comparison. Examples of enhanced speech for baselines and our proposed approach are provided online⁴.

V. RESULTS AND ANALYSIS

In this evaluation, perceptual evaluation of speech quality (PESQ) [60], extended short-time objective intelligibility (ESTOI) [61] and SDR [62] are chosen as the major objective metrics to evaluate the enhancement performance of different models. PESQ is used to evaluate the speech quality, whose values range from -0.5 to 4.5 . ESTOI is the extended version of STOI [63], where the mutual independence assumption among frequency bands are canceled. It ranges from 0 to 1

and is used to indicate speech intelligibility. SDR is widely adopted in blind speech separation and also applies for speech quality evaluation. For three metrics, the higher the scores are, the better the performance is.

Now we compare the performance of the proposed method with the baselines, whose results are presented in Table II and Table III. The best scores are highlighted with boldface in each case. From the tables, one can find that for both seen and unseen speaker cases, the baselines and our proposed method can effectively improve the objective scores under different conditions, which shows that given adequate noises and speakers, the trained model can generalize well toward different noises and speakers, leading to noise and speaker-independent models. For illustration convenience, we will focus on the analysis toward the unseen speaker condition in the following content.

A. Model Comparison Among T-F Domain Based Approaches

When it terms to the comparison among T-F domain based approaches. One can have the following observations.

1) *Phase is significant for speech recovery*: Take CRN and GCRN as an example, we find that average 0.24 score improvements in PESQ are made. It shows that when the

³We reimplement the model following the code <https://github.com/huyanxin/DeepComplexCRN>

⁴We provide examples of enhanced speech processed by different models online at https://github.com/Andong-Li-speech/CTS_TASLP_demos

TABLE III
OBJECTIVE RESULT COMPARISONS AMONG DIFFERENT MODELS IN TERMS OF PESQ, ESTOI AND SDR FOR BABBLE AND FACTORY1 NOISES IN THE UNSEEN SPEAKER TEST SET.

	Metrics SNR(dB)	Cau.	PESQ						ESTOI(%)						SDR(dB)					
			-6	-3	0	3	6	Avg.	-6	-3	0	3	6	Avg.	-6	-3	0	3	6	Avg.
Babble Noise	Noisy	-	1.49	1.64	1.82	2.01	2.23	1.84	24.33	31.51	39.66	48.21	57.74	40.29	-5.89	-2.93	0.05	3.03	6.03	0.06
	LSTM	✓	1.72	1.98	2.25	2.46	2.67	2.22	41.97	52.33	61.63	68.89	75.60	60.09	0.45	3.42	5.93	8.21	10.26	5.65
	CRN	✓	1.67	2.01	2.31	2.56	2.80	2.27	42.63	53.82	63.48	71.20	78.01	61.83	0.53	3.81	6.53	8.98	11.41	6.25
	GCRN	✓	1.88	2.25	2.58	2.83	3.05	2.52	48.41	59.77	69.09	75.78	80.87	66.78	2.77	5.89	8.43	10.47	12.24	7.96
	DCCRN	✓	1.84	2.21	2.55	2.81	3.07	2.49	45.86	57.71	67.61	75.39	81.51	65.62	2.32	5.62	8.49	10.98	13.45	8.17
	ConvTasNet	✓	1.89	2.21	2.50	2.73	2.94	2.45	53.87	64.37	72.61	78.63	83.49	70.59	4.39	7.29	9.69	11.94	13.99	9.46
	ME-Net(Pro.)	✓	1.86	2.19	2.48	2.71	2.93	2.43	47.46	58.46	67.43	74.47	80.38	65.64	1.76	4.98	7.51	9.88	12.24	7.27
	CTS-Net(Pro.)	✓	2.10	2.51	2.83	3.06	3.26	2.75	56.75	68.57	76.35	81.89	85.88	73.89	4.86	8.53	11.04	13.12	15.01	10.51
	BLSTM	✗	2.00	2.30	2.58	2.79	2.99	2.53	53.45	63.26	71.51	77.17	85.02	69.48	2.25	4.92	7.29	9.41	11.48	7.07
	BCRN	✗	1.83	2.25	2.55	2.77	2.97	2.47	48.35	61.51	70.71	76.73	81.98	67.85	1.18	4.73	7.42	9.76	12.08	7.03
	GRN	✗	1.77	2.08	2.32	2.54	2.72	2.29	46.20	56.52	65.57	72.66	78.85	63.96	1.24	4.21	6.76	9.18	11.62	6.60
	DCN	✗	1.70	2.03	2.32	2.57	2.79	2.28	42.19	53.19	63.17	70.81	77.49	61.37	0.74	3.92	6.71	9.14	11.56	6.42
	AECNN	✗	1.85	2.24	2.59	2.86	3.10	2.53	50.75	62.64	72.11	78.37	83.57	69.49	4.07	6.26	9.90	12.12	14.20	9.31
	BGCRN	✗	2.15	2.55	2.85	3.08	3.27	2.78	55.93	68.41	76.68	82.16	86.38	73.91	4.60	7.77	10.42	12.55	14.60	9.99
Factory1 Noise	ME-Net(Pro.)	✗	1.93	2.27	2.56	2.79	3.00	2.51	51.37	62.12	70.53	76.75	82.12	68.58	2.06	5.22	7.79	10.14	12.50	7.54
	CTS-Net(Pro.)	✗	2.27	2.68	3.00	3.22	3.39	2.91	62.48	73.11	80.10	84.49	87.76	77.59	5.94	9.34	11.74	13.70	15.51	11.24
	Noisy	-	1.36	1.55	1.75	1.96	2.17	1.76	23.67	31.97	41.13	50.32	59.78	41.37	-5.90	-2.92	0.04	3.04	6.03	0.06
	LSTM	✓	1.83	2.12	2.36	2.56	2.73	2.32	42.13	53.50	62.75	70.09	75.89	60.87	2.43	4.98	7.19	9.16	10.94	6.94
	CRN	✓	1.84	2.16	2.42	2.66	2.86	2.39	42.39	54.21	64.03	71.92	78.18	62.15	2.36	5.17	7.62	9.88	12.05	7.42
	GCRN	✓	2.00	2.39	2.68	2.90	3.09	2.61	45.73	59.37	68.85	75.70	80.52	66.03	3.90	6.85	9.04	10.83	12.44	8.61
	DCCRN	✓	2.07	2.42	2.70	2.93	3.13	2.65	46.73	59.50	68.92	76.16	81.81	66.62	4.57	7.40	9.75	11.90	13.92	9.51
	ConvTasNet	✓	2.02	2.32	2.56	2.79	2.99	2.54	51.72	63.48	72.07	78.36	82.97	69.72	5.75	8.28	10.40	12.38	14.25	10.21
	ME-Net(Pro.)	✓	2.03	2.31	2.55	2.78	2.97	2.53	47.35	58.49	67.34	74.67	80.26	65.62	3.42	5.98	8.26	10.50	12.73	8.18
	CTS-Net(Pro.)	✓	2.29	2.62	2.86	3.09	3.25	2.82	56.50	67.90	75.49	81.25	85.12	73.25	6.66	9.31	11.34	13.25	15.02	11.12
	BLSTM	✗	2.17	2.43	2.65	2.86	3.04	2.63	53.38	64.51	71.81	77.68	82.01	70.08	3.71	5.96	8.04	10.03	11.91	7.93
	BCRN	✗	2.08	2.40	2.63	2.84	3.01	2.59	51.14	62.72	70.76	77.02	81.81	68.69	3.12	5.90	8.20	10.39	12.57	8.04
	GRN	✗	1.97	2.24	2.44	2.61	2.75	2.40	49.23	59.86	68.38	74.89	80.01	66.48	3.19	5.78	8.08	10.31	12.46	7.96
	DCN	✗	1.95	2.26	2.51	2.73	2.90	2.47	44.99	56.89	66.11	73.40	78.94	64.06	2.99	5.67	8.05	10.31	12.42	7.89
	AECNN	✗	2.03	2.40	2.69	2.94	3.13	2.64	48.79	62.09	71.23	78.00	82.54	68.73	5.70	8.30	10.42	12.35	14.14	10.18
	BGCRN	✗	2.31	2.65	2.93	3.14	3.30	2.87	53.18	66.94	76.06	81.90	85.77	72.77	5.37	8.19	10.54	12.61	14.42	10.22
	ME-Net(Pro.)	✗	2.11	2.40	2.63	2.85	3.05	2.61	51.97	62.52	70.60	77.03	81.92	68.81	3.67	6.20	8.49	10.73	12.95	8.41
	CTS-Net(Pro.)	✗	2.45	2.78	3.02	3.22	3.37	2.97	62.58	72.72	78.33	83.95	87.07	76.93	7.51	9.92	11.92	13.68	15.35	11.67

phase is incorporated, better speech quality can be obtained. A similar tendency is also observed for ESTOI and SDR.

2) *The utilization of future information is conducive to objective performance:* For example, when uni-directional LSTM layers in the causal systems are replaced by the bi-directional versions, notable score improvements are made. This is because when the utterance-level information is utilized, the feature in the current timestep can be better represented by the information from both directions (past and future).

3) *Performance bottleneck for complex spectral mapping approaches with single stage:* Take GCRN and DCCRN as an example, one can find that both methods have close scores under different conditions. This is mainly because both of them have the similar network topology, and the same loss is adopted to estimate magnitude and phase information. It also reveals the fact that current complex spectral mapping approaches with the single-stage paradigm have suffered from the performance bottleneck.

4) *CTS-Net consistently outperforms the previous models in the metric scores:* For example, when babble noise is given for causal settings, compared with GCRN, CTS-Net achieves 0.23, 7.11%, and 2.55dB score improvements on average in terms of PESQ, ESTOI, and SDR, respectively. For that of factory1 noise, the improvements are 0.21, 7.22% and

2.51dB, respectively. This indicates the notable advantage of two-stage networks over single-stage networks in the complex spectral mapping. We argue that for single-stage topology, despite the multi-task learning for RI and magnitude can be leveraged to guarantee the magnitude and phase optimization, the estimation output is enforced to compromise between the two targets, limiting the performance when the optimization problem is relatively difficult, *e.g.*, low SNRs. For the proposed method, as two targets are decoupled into two networks and avoid the direct optimization trade-off within one stage, *i.e.*, ME-Net only aims to recover the magnitude while CS-Net attempts to modify the phase, better solutions can be obtained. It is interesting to find that in [33], a similar conclusion is also drawn, where ConvTasNet serves as the post-processing module to further refine the speech estimated by the previous T-F domain based network, leading to significant improvements in objective metrics. It shows the potential of the multi-stage pipeline to surpass existing single-network schemes.

B. CTS-Net vs. Time-domain Systems

We then compare between CTS-Net and time-domain based systems. One can find that in causal case, for ConvTasNet and CTS-Net, our system enjoys sufficient advantage, *e.g.*, around 0.29, 3.42%, and 0.98dB average improvements in PESQ,

TABLE IV
ABLATION STUDY WITH RESPECT TO TCM TYPE AND THE GATED CONVOLUTIONS IN BOTH ENCODER AND DECODER. THE VALUES ARE AVERAGED OVER ALL THE SNRS IN THE UNSEEN SPEAKER TEST SET.

Models	id	Gate	S-TCM	TCM	Babble		Factory1	
					PESQ	ESTOI(%)	PESQ	ESTOI(%)
Noisy	-	-	-	-	1.84	41.44	1.76	41.37
ME-Net	1	✓	✗	✗	2.20	58.93	2.35	60.19
ME-Net	2	✗	✓	✗	2.39	64.12	2.48	64.07
ME-Net	3	✓	✗	✓	2.41	65.05	2.49	64.65
ME-Net	4	✓	✓	✗	2.43	65.64	2.53	65.62

ESTOI, and SDR are observed. In addition, even for causal setup, CTS-Net still yields average 0.20, 4.46%, and 1.07dB score improvements over AECNN, and the performance gap can be further enlarged for non-causal case. From the results above, one can conclude that our two-stage paradigm in the T-F domain enjoys significant performance superiority over previous time-domain systems.

C. ME-Net vs. CTS-Net

We compare between ME-Net and CTS-Net to reveal the advantages of the second stage. Take babble noise as an example, when causal setup is given, going from ME-Net to CTS-Net, we observe dramatic improvements in all three objective metrics, *e.g.*, average 0.32, 8.25%, and 3.24dB increase in PESQ, ESTOI, and SDR scores are achieved. It shows that when the second stage is applied, both speech quality and intelligibility can be notably improved. The reasons can be attributed as two-fold. For one thing, after the first stage, there still exist some residual noises, and some detailed information are also lost. The second stage actually serves as the post-processing module to further suppress the remaining interference, and repair the spectral detail. For another, the second stage can implicitly recover the phase information by modifying real and imaginary spectra, which helps to improve the speech quality.

D. Effects of Gating Mechanisms and TCMs

Table IV presents the effects of gating mechanisms in both encoder and decoder, and sequential modeling units. As ME-Net and CS-Net have the similar network structure, we only conduct the experiments on ME-Net, and similar conclusions can also apply to CS-Net. The values are averaged over different SNRs and the best results are highlighted with boldface in each case. From the table, one can obtain the following observations.

1) *Temporal sequential modeling is significant for speech recovery*: For example, going from ME-Net with id-1 to that of id-4, average 0.23 and 0.18 PESQ improvements are observed while 6.71% and 5.43% ESTOI improvements are achieved for babble and factory1 cases, respectively. It shows that when stacked sequential modeling units are introduced, obvious improvements in speech quality and intelligibility can be made. This is because there exist both short and long-range relation between frames. Despite encoder and decoder can grasp the local relation with multiple 2-D convolutions,

they neglect the long-range contextual information, which inevitably causes performance degradation.

2) *Gating mechanisms can facilitate metric performance*: For example, going from id-2 to id-4, average 0.04 and 1.52% improvements are achieved in PESQ and ESTOI for babble noise. This is because the gated convolutions can effectively modulate the distribution of features in both the encoder and decoder, and meanwhile, facilitate the information flow during the gradient propagation.

3) *TCM vs. S-TCM*: Going from TCM to S-TCM, we observe further improvements in both PESQ and ESTOI. We argue that despite in TCM, the features are projected into a higher dimension space, compared with depthwise dilated convolutions, regular dilated convolutions have the better capability in feature transformation and sequential learning from different temporal scales. In addition, the spectrum has a rather sparse distribution in the T-F domain and it is de facto adequate to learn with a lower dimension instead of a higher feature space.

E. Illustration toward the Role of the Second Stage

In this part, we demonstrate the role of the second stage in speech recovery in detail. Three metrics are adopted, namely SNR, log-spectral distance (LSD) [64] and phase distance (PD) [65]. LSD is adopted to measure the spectral distortion between clean and enhanced spectra, whose calculation formula is given as:

$$\text{LSD}(S, \tilde{S}) = \frac{1}{L} \sum_{l=0}^{L-1} \left[\frac{2}{N} \sum_{k=1}^{K/2} \left(\sigma S_{k,l} - \sigma \tilde{S}_{k,l} \right)^2 \right]^{\frac{1}{2}}, \quad (22)$$

where $\sigma S_{k,l} \triangleq \max\{20 \log_{10}|S_{k,l}|, \tau\}$ is the log-power spectrum of $S_{k,l}$, and τ is a small value to scale the value range. In this study, we set $\tau = \max_{k,l}\{20 \log_{10}|S_{k,l}|\} - 50$ to confine the dynamic range into 50dB. Note that voice activity detection (VAD) is adopted to remove the silence regions before calculation.

PD is considered in [65] to evaluate the phase recovery performance, where the phase difference in each T-F bin is weighted by the corresponding magnitude value of target spectrum to emphasize the relative importance of different spectral regions. The calculation formula is given as:

$$\text{PD}(S, \tilde{S}) = \sum_{k,l} \frac{|S_{k,l}|}{\sum_{k',l'} |S_{k',l'}|} \angle \left(S_{k,l}, \tilde{S}_{k,l} \right), \quad (23)$$

where $\angle \left(S_{k,l}, \tilde{S}_{k,l} \right)$ denotes the angle difference between $S_{k,l}$ and $\tilde{S}_{k,l}$ with the range of $[0^\circ, 180^\circ]$.

In this part, we re-generate the test samples from -6dB to 6dB with 1dB interval under factory1 noise, and five models are measured, namely GCRN, AECNN, ConvTasNet, ME-Net (causal) and CTS-Net (causal). For each input SNR, metric improvements over noisy speeches in terms of SNR, LSD and PD are measured, *i.e.*, iSNR (SNR improvement), rLSD (LSD reduction), rPD (PD reduction), whose definitions are given as:

$$\text{iSNR} = \text{SNR} \left(S, \tilde{S} \right) - \text{SNR} \left(S, X \right), \quad (24)$$

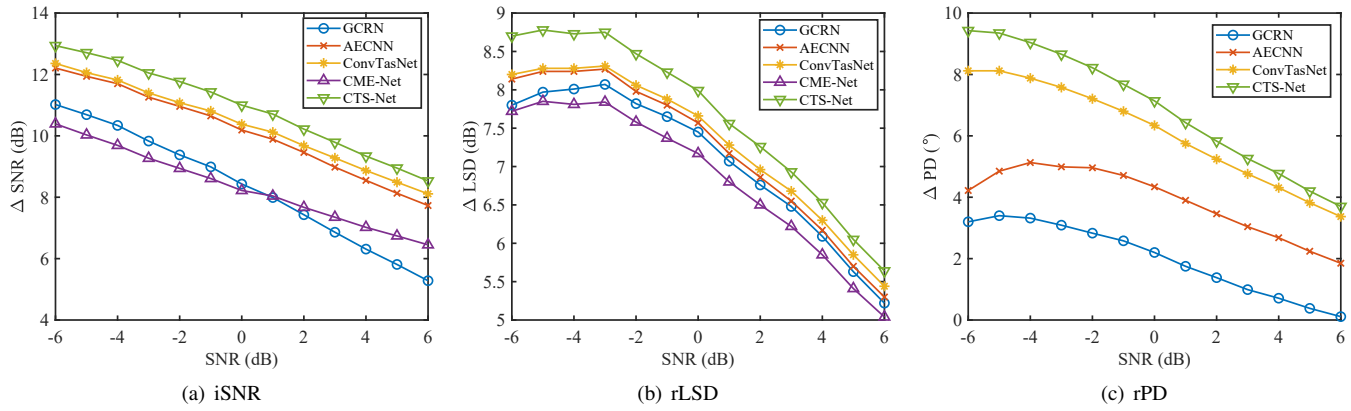


Fig. 3. iSNR, rLSD and rPD under different noisy conditions for different models. The values in each input SNR are averaged with factory1 noise from the test set. The input SNR value ranges from -6dB to 6dB with 1dB interval. (a): the SNR improvements. (b): the LSD reduction. (c): the PD reduction.

TABLE V
RESULTS COMPARISON WITH REF1 AND REF2. THE VALUES ARE
AVERAGED OVER ALL THE SNRS.

Models	Babble			Factory1		
	PESQ	ESTOI(%)	SDR(dB)	PESQ	ESTOI(%)	SDR(dB)
Ref1	2.73	72.74	9.87	2.80	71.94	10.46
Ref2	2.75	73.14	10.04	2.81	72.77	10.66
CTS-Net	2.78	74.73	10.46	2.83	73.89	10.98

$$\text{rLSD} = \text{LSD}(S, X) - \text{LSD}(S, \tilde{S}), \quad (25)$$

$$\text{rPD} = \text{PD}(S, X) - \text{PD}(S, \tilde{S}), \quad (26)$$

The results are shown in Fig. 3, where the curves of different models are plotted with different formats. Note that as ME-Net only estimates the magnitude of the spectrum without phase modification, rPD remains zero for different cases, which is not given in Fig. 3 (c) for depiction convenience.

From the figures, one can obtain the following observations. Firstly, CTS-Net consistently outperforms another three base-lines in terms of three metrics, which shows the superiority of CTS-Net in both spectral refinement and phase modification. Secondly, switching from ME-Net to CTS-Net, we observe notable performance improvements in the three metrics. For example, for SNR and LSD, when the second stage is adopted, better performance in SNR improvements and LSD decrease is observed, indicating that the second stage helps to further suppress the remaining noises of the first stage, while the spectral components can also get further polished. For PD, while ME-Net does not change the phase, the second network obtains effective phase modification, which also reveals the importance to decouple magnitude and phase estimations.

F. Advantage of the Proposed Training Strategy

To reveal the advantage of our two-stage training strategy, we set two reference groups called Ref1 and Ref2. Ref1 has a single-stage network topology similar to CS-Net but we add more channels and the total number of parameters is around 1.5 times of CTS-Net. Ref2 has the same architecture of CTS-Net except we directly optimize the whole network from scratch without pre-training. The comparison results among

the three models are given in Table V. Two observations can be made. Firstly, going from Ref1 to Ref2, consistent improvements are observed, which verifies the necessity of two-stage architecture. Besides, compared with training from scratch, when the pre-training is adopted in the first stage, better performance can be obtained, which emphasizes the importance of pre-training in multi-stage optimization.

G. Effect of Global Residual Connection in the Second Stage

At this point, one may ask one question, *why we introduce a global residual connection in the second stage instead of directly estimating the complex spectrum*. To investigate its effectiveness, we consider two implementations for the second stage after the coarse spectrum $\tilde{S}^{me}$ is obtained from the first stage. One is to directly estimate the clean complex spectrum while the other adopts our proposed residual solution, *i.e.*,

$$\tilde{S}^{cs1} = \mathcal{G}_{cs1}(\tilde{S}^{me}, X), \quad (27)$$

$$\tilde{S}^{cs2} = \tilde{S}^{me} + \mathcal{G}_{cs2}(\tilde{S}^{me}, X). \quad (28)$$

During the training process, for the first $\tilde{S}^{cs1}$, the network attempts to decompose the input features at the encoder and then reconstruct the clean spectrum at the decoder from scratch. As the network is initialized with random values, the training loss tends to be large at the first several epochs. For the second $\tilde{S}^{cs2}$, however, as the network only needs to complement the residual parts based on the prior estimation from the last stage, it inherits the previous stage result, which leads to a small training loss at the beginning. So we conjecture that a global residual connection is conducive to network convergence.

To investigate the performance improvement brought by the second stage, it is necessary and significant to discuss the upper and lower performance bound of the two implementations. According to [66], it is much easier for a network with global residual connection to learn the identity mapping than that of no residual connection in the practical learning process. When the network is powerful enough to mapping the target theoretically, both two implementations can perfectly map the target, *i.e.*, $\mathcal{G}_{cs1}(\tilde{S}^{me}, X) = S$, and $\mathcal{G}_{cs2}(\tilde{S}^{me}, X) = S - \tilde{S}^{me}$. It means that both two types have the same upper

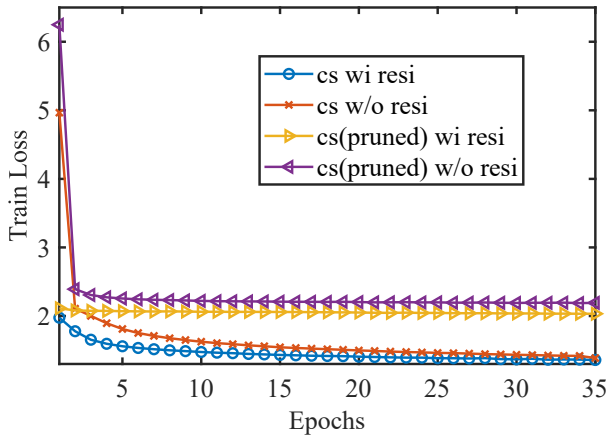


Fig. 4. Loss curves of the second stage in terms of whether to use global residual connection and different number of parameters. “cs” denotes CS-Net. “wi resi” denotes using a global residual connection, and “w/o” denotes no global residual connection is adopted.

TABLE VI

OBJECTIVE EVALUATIONS WITH RESPECT TO WHETHER TO USE GLOBAL RESIDUAL CONNECTION AND THE NUMBER OF NETWORK PARAMETERS. THE SCORES ARE AVERAGED OVER THE TEST SET WITH FACTORY1 NOISE.

Metrics	PESQ				ESTOI(%)			
	SNR(dB)	-6	0	6	Avg.	-6	0	6
no prune	cs wi resi	2.21	2.79	3.19	2.73	54.73	73.92	84.17
	cs w/o resi	2.20	2.78	3.18	2.72	54.24	73.56	83.94
pruned	cs wi resi	2.11	2.65	3.06	2.61	48.65	69.13	81.35
	cs w/o resi	2.05	2.58	3.01	2.54	47.56	67.69	80.44

performance bound, *i.e.*, $\sup\{\tilde{S}^{cs1}\} = \sup\{\tilde{S}^{cs2}\} = S$. At the other extreme, when fewer parameters is given in the second stage network, due to the weak mapping capability of the second stage, it is quite difficult for the first implementation to learn and reconstruct a refined spectrum from scratch, not to mention the identity mapping as the previous estimation. For the second type, however, as a global residual connection is introduced, it can easily obtain an identity mapping and further refine the spectrum with the improvement of the mapping capability of the network. As a consequence, a network with global residual connection can better inherit the result from the previous stage and exhibit better lower performance bound than that of no residual connection in the second stage, *i.e.*, $\inf\{\tilde{S}^{cs2}\} > \inf\{\tilde{S}^{cs1}\}$.

To verify the point, we set two types of networks in the second stage. For the first type, we keep the same number of parameters to CS-Net to obtain a powerful network. For the second type, we prune the network parameters until around 3% parameters of CS-Net remain. 30,000 pairs are randomly selected from the training set to train both networks. To avoid the gradient influence from the first stage, we freeze the pre-trained weights of the first stage and only train the second network. The loss curves over different situations are shown in Fig. 4. One can find that no matter whether the pruning is adopted, when the residual connection is given, the network

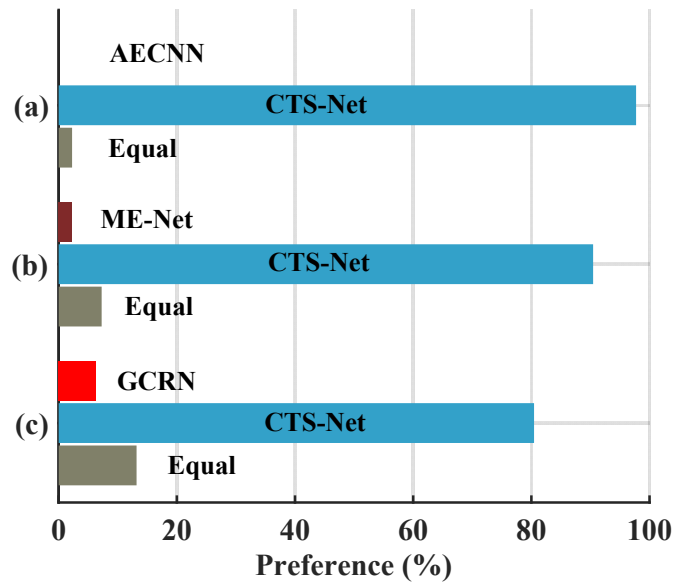


Fig. 5. Subjective evaluations with AB test. (a): subjective comparison between AECNN and CTS-Net. (b): subjective comparison between ME-Net and CTS-Net. (c): subjective comparison between GCRN and CTS-Net.

achieves a consistently faster convergence speed than that of no residual connection.

We also report the performance scores with respect to global residual connection and the number of parameters, as shown in Table VI. One can find that no matter whether the network is pruned, when the residual connection is adopted, consistently better performance can be made in both PESQ and ESTOI. In addition, when the network is not pruned, the performance between “wi resi” (with residual connection) and “w/o resi” (without residual connection) are close. However, if the number of parameters is heavily pruned, we observe notable advantages of residual estimation over direct estimation, which demonstrates that an external residual connection can indeed better preserve the performance of the previous stage.

H. Subjective Evaluation

We also conduct the subjective AB test to validate the advantage of our proposed approach over the baselines in the subjective quality⁵. 11 volunteers participated in the test. Three groups are set, namely AECNN vs CTS-Net (group-a), ME-Net vs CTS-Net (group-b), and GCRN vs CTS-Net (group-c). Each group consists of 20 utterance pairs processed by two algorithms. All the utterances were randomly selected from the test set and chunked to 4 seconds. For all the participants, they were placed at a quiet office to finish the listening test. All the utterances were placed by a PC and presented over Sennheiser HD 380 Pro (Wedemark, German). The algorithm labels in each pair were randomly shuffled to avoid subjective inertia. For listeners, they would first hear the original noisy speech, followed by two processed speeches. Later, they were asked to select the speech label with overall better quality. In

⁵We provide the subjective testing platform in https://github.com/Andong-Li-speech/AB_test_platform_TASLP

TABLE VII

COMPARISONS IN TERMS OF MODEL SIZE (IN MILLION), MACs (IN MILLION), PROCESSING TIME PER SECOND ON CPU, MEMORY FOOTPRINT (IN GB), AND TRAINING BATCH TIME (IN SECOND).

Models	Model size(M)	MACs(G/s)	Process time(s)	Memory footprint(GB)		Training batch time(s)
				Forward	Backward	
LSTM	30.22	3.72	0.46	0.26	0.64	0.10
CRN	17.58	2.57	0.27	0.18	0.33	0.07
GCRN	9.77	2.42	0.17	0.19	0.22	0.15
DCCRN	3.67	11.13	0.40	0.66	0.67	0.27
ConvTasNet	5.00	5.23	0.71	2.11	2.13	0.44
ME-Net(Pro.)	1.96	2.01	0.13	0.25	0.26	0.29
CTS-Net(Pro.)	4.35	5.57	0.34	0.59	0.60	0.63

TABLE VIII

COMPARISON WITH OTHER STATE-OF-THE-ART SYSTEMS ON DNS CHALLENGE DATASET.

Models	Cau.	Reverb				No Reverb			
		WB-PESQ	PESQ	STOI(%)	SI-SDR(dB)	WB-PESQ	PESQ	STOI(%)	SI-SDR(dB)
Noisy	-	1.82	2.75	86.62	9.03	1.58	2.45	91.52	9.07
NSNet [68]	✓	2.37	3.08	90.43	14.72	2.15	2.87	94.47	15.61
DTLN [69]	✓	-	2.70	84.68	10.53	-	3.04	94.76	16.34
STFT-TCN [70]	✗	2.75	-	-	-	2.73	-	-	-
DCCRN [56]	✓	-	3.22	-	-	-	3.27	-	-
PoCoNet [71]	✗	2.83	-	-	-	2.75	-	-	-
ME-Net(Pro.)	✓	2.91	3.43	92.57	15.33	2.76	3.31	96.21	16.69
CTS-Net(Pro.)	✓	3.02	3.47	92.70	15.58	2.94	3.42	96.66	17.99

addition, “Equal” option was also provided if they couldnot distinguish between two speeches.

Fig. 5 gives the testing results. Several observations can be made. Firstly, we are surprised at first to find that nearly all the preference is given to CTS-Net in the group-a despite the fact that decent performance has been made for AECNN in the objective metrics. After hearing the samples, we attribute the reason as a strange humming-alike artifact is introduced to the speech processed by AECNN, which heavily hampers the subjective quality. A similar phenomenon is also reported in [67]. Secondly, compared with ME-Net, after the second stage is applied, notable preference is given, which indicates that the second stage is crucial for subjective quality improvement. Thirdly, although a complex-spectral mapping approach is adopted for both GCRN and CTS-Net, the latter enjoys consistently better preference in speech quality, which emphasizes the importance of the two-stage topology.

I. Model Complexity for Real-Time Processing

For causal SE systems, we need to take several statistics into account in real scenarios, *e.g.*, model size, the number of multiply-accumulate operations (MACs) per second, processing time, memory footprint, and training batch time. To this end, Table VII presents the results of seven causal SE systems in terms of different factors. Note that we calculate the processing time with an input audio of 16000 samples (1 second) for all models on an Intel Core(TM) i5-4300 CPU clocked at 1.90GHz. The memory footprint and training batch time are evaluated with a batch size of 4 on NVIDIA GeForce GTX 1660Ti. One can find that ME-Net has the smallest value in the first three metrics. When the two-stage topology is

TABLE IX

COMPARISON WITH OTHER STATE-OF-THE-ART SYSTEMS ON VOICE BANK + DEMAND DATASET.

Model	WB-PESQ	CSIG	CBAK	COVL
Noisy	1.97	3.35	2.44	2.63
SEGAN [22]	2.16	3.48	2.94	2.80
MMSE-GAN [72]	2.53	3.80	3.12	3.14
MetricGAN [73]	2.86	3.99	3.18	3.42
Wavenet [74]	-	3.62	3.23	2.98
MDPhD [75]	2.70	3.85	3.39	3.27
STFT-TCN [70]	2.89	4.24	3.40	3.56
M-CRAN-MSE [76]	2.92	4.16	3.24	3.54
TFT-Net [77]	2.75	3.93	3.44	3.34
ME-Net(Pro.)	2.90	4.25	3.42	3.59
CTS-Net(Pro.)	2.92	4.25	3.46	3.59

introduced, it inevitably introduces more parameters, computation consumption, and processing delay. However, compared with ConvTasNet, CTS-Net is still more advantageous in the processing delay and memory footprint. This is mainly because ConvTasNet utilizes an extremely small frame length (*e.g.*, 2ms) over that in the previous literature (*e.g.*, 20ms, 32ms), and more frames are needed to process, which increases the overall processing time and memory consumption.

J. Comparison with Other State-of-the-art Methods

Except for the WSJ0-SI84 corpus, we also make the comparison with other state-of-the-art approaches on two benchmarks, namely DNS Challenge dataset [68] and Voice Bank + Demand [78]. DNS Challenge dataset provides over 500 hours of clean clips from 2150 speakers and over 180 hours of noise clips. For model evaluation, it provides a validation set of two

categories, namely with reverberation and no reverberation, each of which has 150 noisy-clean clips with SNRs distributed between 0dB to 20dB. Voice Bank + Demand includes 11,572 utterances for training. 824 utterances are used for testing under four SNRs {2.5dB, 7.5dB, 12.5dB, and 17.5dB}.

We totally generate around 3000 hours noisy-clean pairs for training and the results on the DNS Challenge test set are presented in Table VIII, where four metrics are measured, namely wide-band PESQ (dubbed WB-PESQ) [79], narrow-band PESQ⁶, STOI, and SI-SDR. From the table, one can observe that CTS-Net yields the highest score in different cases among these advanced systems. Note that DCCRN and PoCoNet ranked top-1 in real-time and non-real-time tracks in the INTERSPEECH 2020 DNS challenge, respectively. Nonetheless, CTS-Net still surpasses them by a large margin.

For Voice Bank + DEMAND corpus, the evaluation results are presented in Table IX, which are based on WB-PESQ and three MOS-based metrics [80], *i.e.*, CSIG, CBAK, and COVL. From the table, one can find that CTS-Net achieves overall better performance in both metrics, which indicates the effectiveness of our method. However, going from ME-Net to CTS-Net, just slight score improvements are achieved, which can be explained from two aspects. For one thing, the SNR level of the test set is rather high, leading to less noise residual left after the first stage. So it is quite difficult for the second stage to further refine the spectrum. For another, the phase difference between noisy and clean speech is relatively small due to the high SNR level, which can only bring paltry benefit after modifying the phase information in the second stage.

VI. DISCUSSIONS

The above objective and subjective test results reveal the effectiveness of our proposed complex spectral mapping based two-stage approach called CTS-Net. Some valuable observations can be made.

Compared with single-stage based baselines, our method achieves consistent advantages in both objective and subjective evaluations. The reasons are two-fold. Firstly, instead of mapping the target within one step, it seems more sensible and easier to decompose the original difficult problem into several sub-problems, which also follows the classic proverb: “two-heads are better than one”. In this way, better solutions can be accomplished for each sub-problem. In this study, instead of estimating magnitude and phase simultaneously within a single stage, our method attempts to convert the multi-objective optimization into the sequential optimization, *i.e.*, we choose the magnitude estimation as the first step, followed by phase modification as the second step. In this way, the pre-estimated magnitude can serve as the prior information to boost the subsequent phase optimization. Secondly, after the first step, some residual noises inevitably exist, and the detailed spectral information tends to get lost due to the network performance bound. Therefore, the second stage actually serves as the post-processing module to further suppress the noise components while effectively repairing the spectrum. Note that a global

residual connection strategy is adopted in the second stage, so the second network only needs to focus on the missing details rather than re-generate the overall spectrum from scratch, which facilitates the convergence speed of the training process.

According to the results from three datasets, we also conclude that if the SNR level is relatively low, where the noise components usually dominate the spectra, it seems rather tough for a single-stage network to extract the speech contents with good quality. So it is crucial to introduce the auxiliary network as the post-processing module to further refine the speech from the previous estimation. At the other extreme, if the SNR level is high enough, it seems limited to introduce the extra assistance since a single-stage network is adequate enough to accomplish the task.

In summary, our proposed complex spectral mapping based two-stage network can effectively extract the speech information from the noisy recordings and also surpass the previous baselines in both objective and subjective evaluations.

VII. CONCLUSION

In this paper, we present a complex spectral mapping based two-stage approach for monaural speech enhancement. In the first stage, a network is tasked with estimating the magnitude of the spectrum. To further polish the spectrum while modifying the phase information, an auxiliary network is adopted with an external residual connection, which significantly improves the speech quality. Besides, we propose a light-weight temporal convolutional module to capture long-range dependencies, which dramatically decreases the trainable parameters while obtaining better performance over the original version. We conduct extensive experiments on the WSJ0-SI84, DNS Challenge dataset, and Voice Bank + DEMAND dataset. The results demonstrate that our method consistently outperforms the previous advanced systems. In the future, we aim to investigate the effectiveness of our approach in more real scenarios where both noise, reverberation, and speaker interference co-exist.

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⁶The utilized PESQ in the previous experiments on WSJ0-SI84 is also based on narrow-band criterion.

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